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Advances and perspectives in REBCO coated conductor production at SuperPower

Yifei Zhang

2025 Coated Conductor for Applications Workshop

CERN

March 11, 2025



International Workshop on Coated Conductors for Applications

CERN | March 11-13, 2025



– SuperPower Inc. is a subsidiary of Furukawa Electric Co. Ltd.

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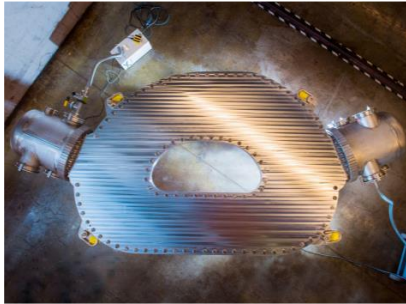
Topics

- Currently focused applications
- Status of REBCO coated conductor manufacturing
- Ongoing manufacturing technology development efforts

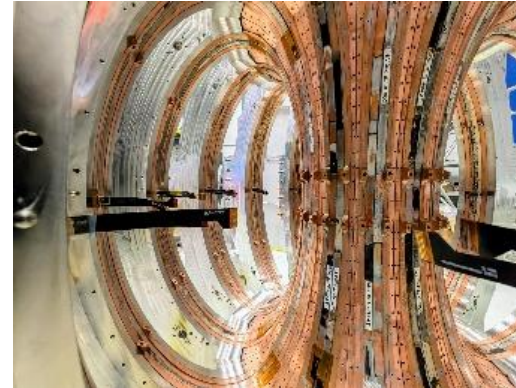
Currently focused applications

- Fusion reactor magnets
- High-field NMR magnets
- High-field scientific research magnets
- High-field accelerator magnets
- Electric machines magnets

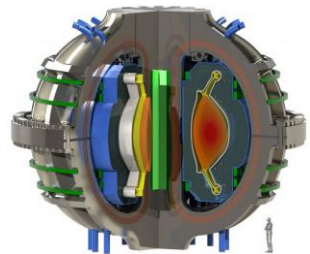
REBCO CC for fusion reactor magnets



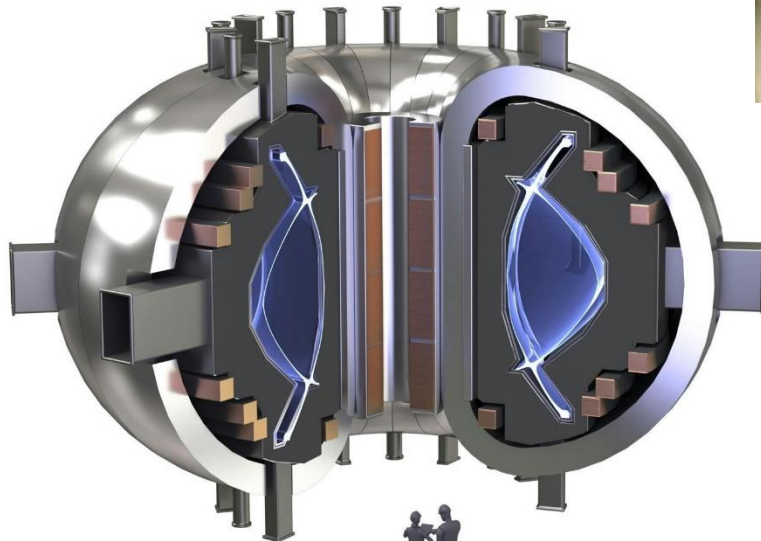
TFMC



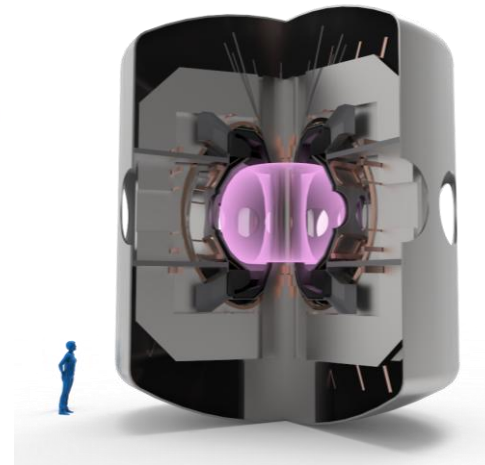
Demo4



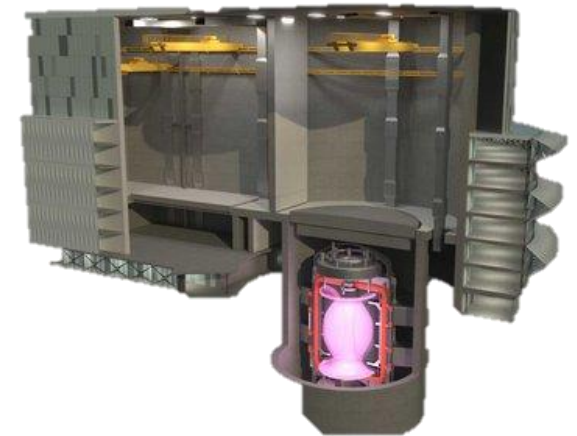
SPARC



ARC

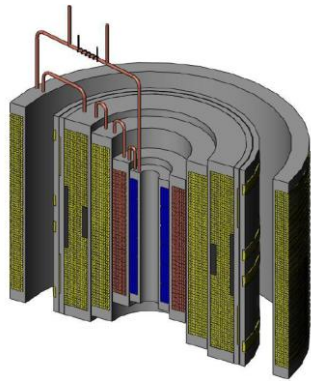


STX

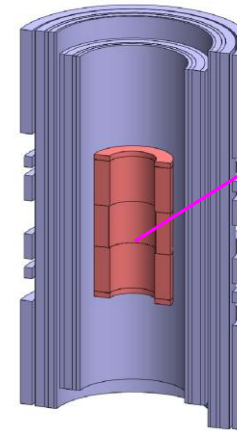


STEP

REBCO CC for high-field NMR magnets



High-field magnets for commercial NMR system



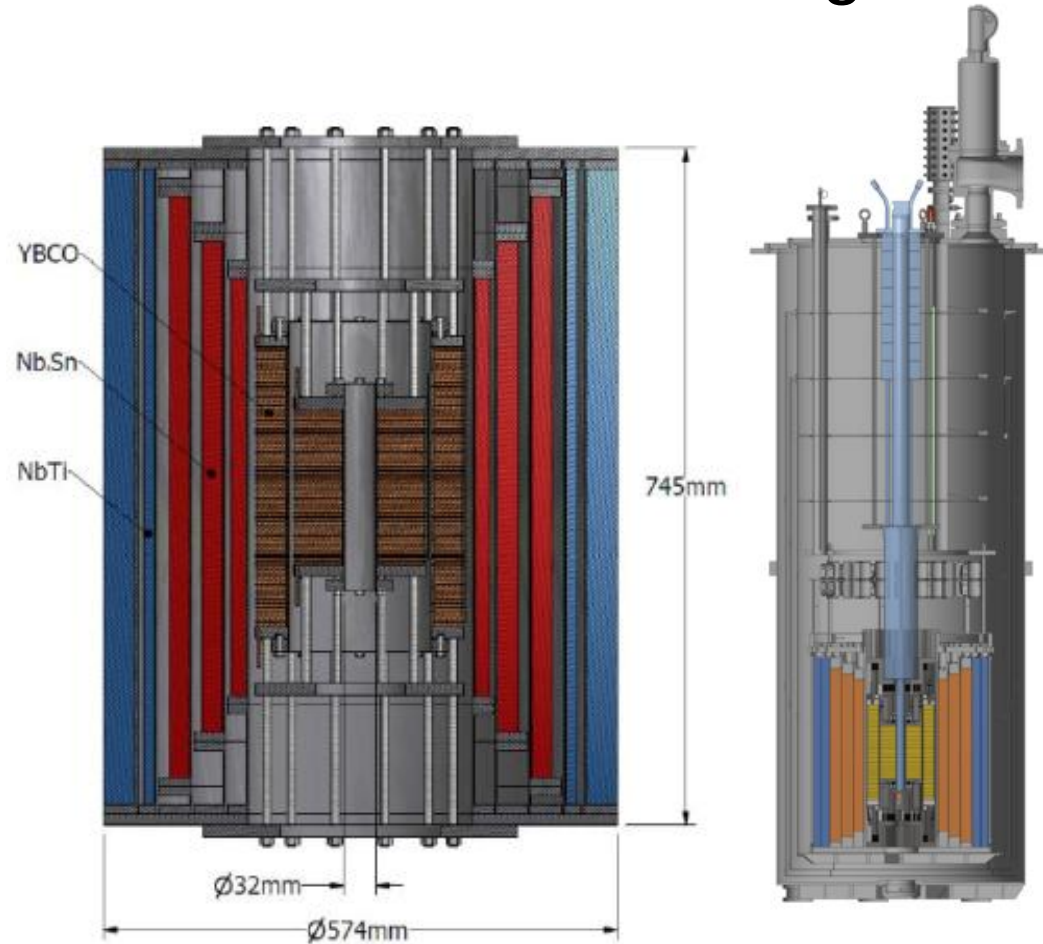
H835



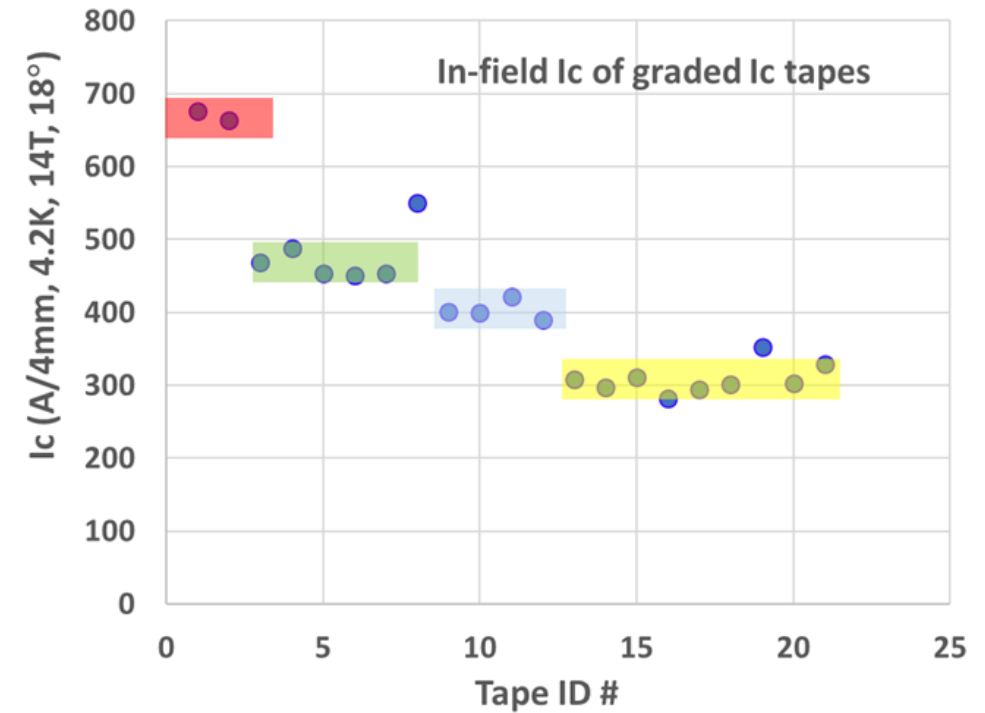
1.3GHz NMR underdevelopment

High-field all-superconductor magnets for NMR

REBCO CC for high-field scientific research magnets



32T user magnet with 17T HTS insert and 15T LTS outsert



Graded I_c REBCO CC for the ongoing 40T magnet project

REBCO CC production at SuperPower

Core manufacturing technologies:

- Ion Beam Assisted Deposition (**IBAD**) to create texture
- Metal Organic Chemical Vapor Deposition (**MOCVD**) with artificial pinning centers (APCs)

REBCO formulation

- **AP** – for applications at various conditions
- **HM** – for applications at lower temperatures and higher fields

Dimensions

- Width = 2, 3, 4, 6, 12 mm
- Piece length = typically ~300 m, and up to 900 m depending on specification

Non-superconducting layer thickness

- Substrate thickness = 30 or 50 μm
- Total Ag thickness = 3~5 μm
- Total Cu stabilizer thickness = 10~120 μm

Critical Current

- $I_c(77\text{K, s.f.})/4\text{mm} = 120\sim 160$ A (AP tapes)
- $I_c(4.2\text{K}, 15\text{T//c})/4\text{mm} = 400\sim 500$ A (HM tapes)
- $I_c(20\text{K, } 20\text{T//c})/4\text{mm} = 160\sim 200$ A (HM tapes)

Customization

- Polyimide insulation (currently by wrapping, polyimide coating coming)
- Soldered lap joints (≤ 100 n Ω -cm²)
- Solder coating (SnPb solder, typically 10 μm total)
- Filamentization (down to 200 μm in filament width)
- AgAu coating instead of pure Ag (for current leads)

REBCO CC production at SuperPower

- Phase 1 capacity expansion completed in 2023
- Production capacity doubled from ~ 120 km/yr to ~ **240** km/yr (12 mm-w)



IBAD system IBAD2



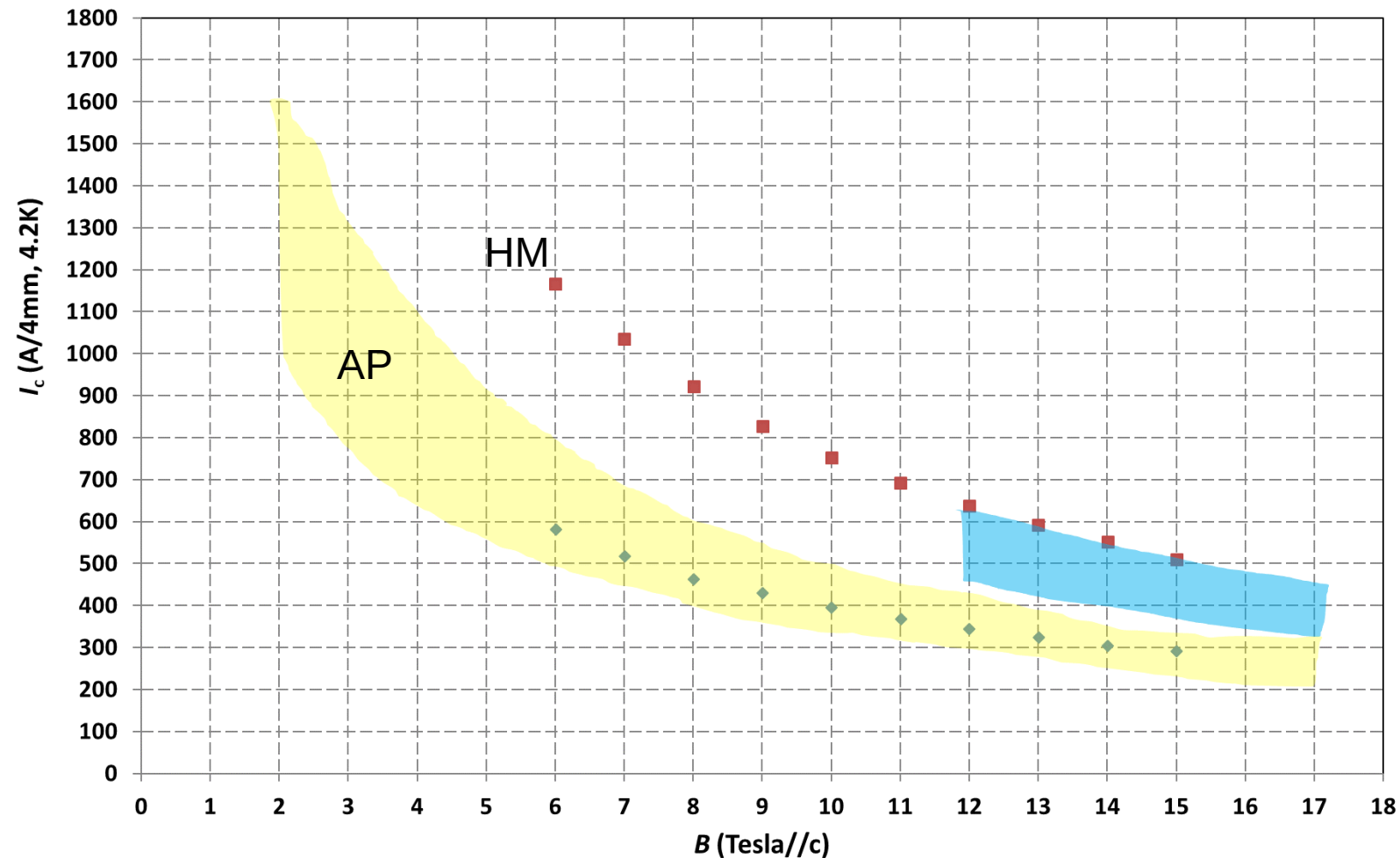
MOCVD system M6



MOCVD system M7

Manufacturing equipment added in Phase 1 capacity expansion

In-field I_c of REBCO CC made with new MOCVD systems



Major ongoing manufacturing technology development efforts

- Increasing yield, throughput, and consistency
- Establishment of QMS compliant with ISO 9001
- Design of next generation MOCVD system
- Development of new REBCO formula specifically for fusion
- Extending in-field I_c testing capability
- Filamentized REBCO CC via laser striation
- Polyimide coating for insulation
- Improvement of slit edge quality by hybrid slitting

Transition from small-scale to large-scale production

- **Small-scale production**

- Low throughput and low yield
- Low material usage efficiency
- Lack of in-line monitoring and control
- Longer machine downtime for maintenance and repair
- High production cost

Transition



- **Large-scale production**

- High throughput and high yield
- Higher material usage efficiency
- Improved operation stability with in-line monitoring, closed-loop control
- Reduced machine downtime (planned and unplanned)
- Reduced production cost

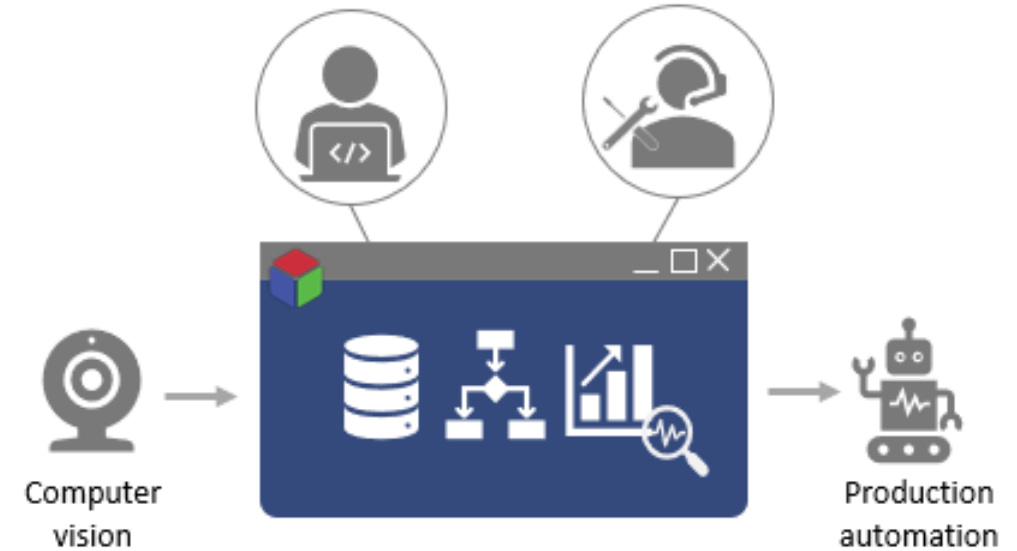


Next generation MOCVD system under design

CTQT (Continuous Tape Quality Tracking) system



- **CTQT** is a Data Warehouse platform
- It includes the data acquisition, analysis and reporting tools,
- On the platform the process and measurement data is stored in an organized manner,
- Engineers, technicians, managers and researchers could access the data anywhere from the intranet (or externally by using a VPN) and visualize it to gain insight into the manufacturing processes.



CTQT is under development by **FETI**, our sister company in Budapest, Hungary

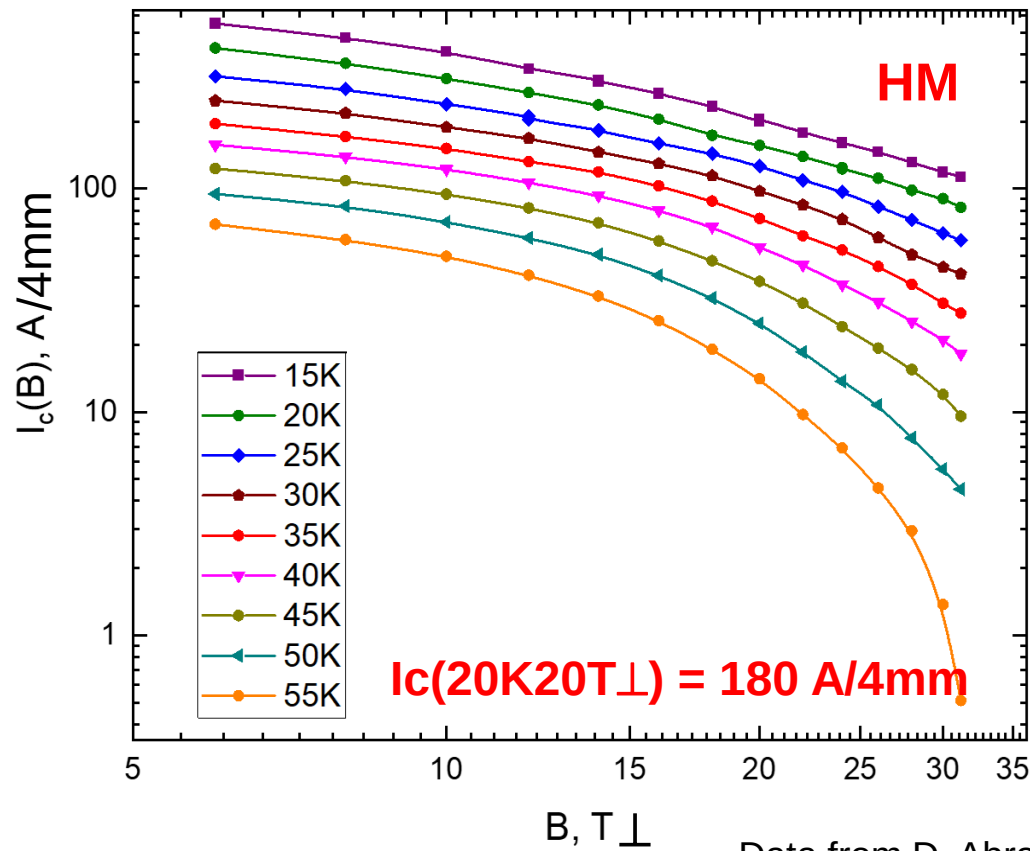
Implementation of ISO 9001:2015

- SuperPower has more than 10 years experiences with key customers in meeting requirements from ISO 9001:2015 (5 times second-party QMS audits)
- Decision has been made to move forward to become a certified company
- Quality team formed and resources put in place
- Gap audit to be carried out by an FEC sister company before having registration audit
- Currently focusing on QMS documentation, employee training, establishing and implementing SOPs and WINs

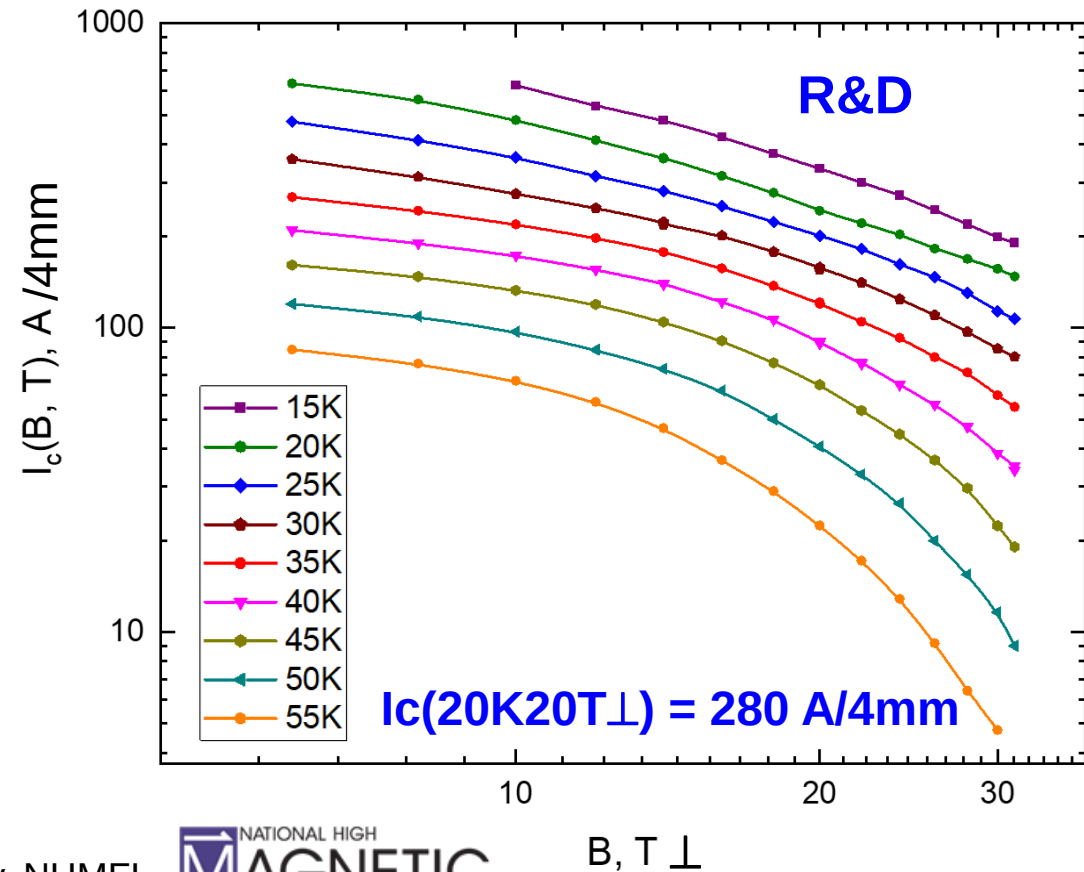


Development of new REBCO formula for fusion

- Further improvement in in-field performance around 20K20T where fusion magnets operate
- New REBCO formula named **FM**, to be launched to the market in 2026



Data from D. Abraimov, NHMFL

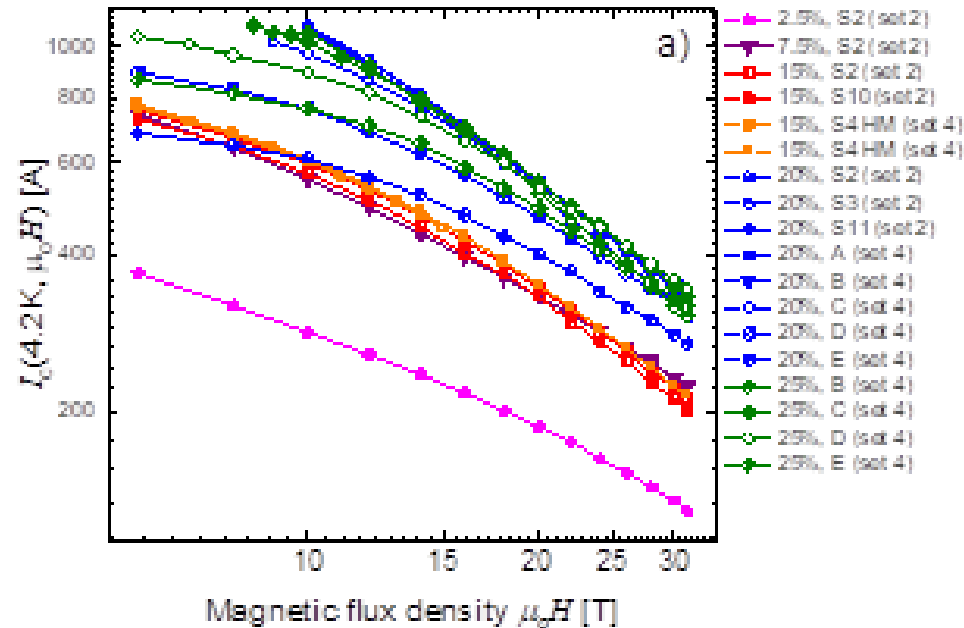


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Extending low-temperature high-field I_c testing capability

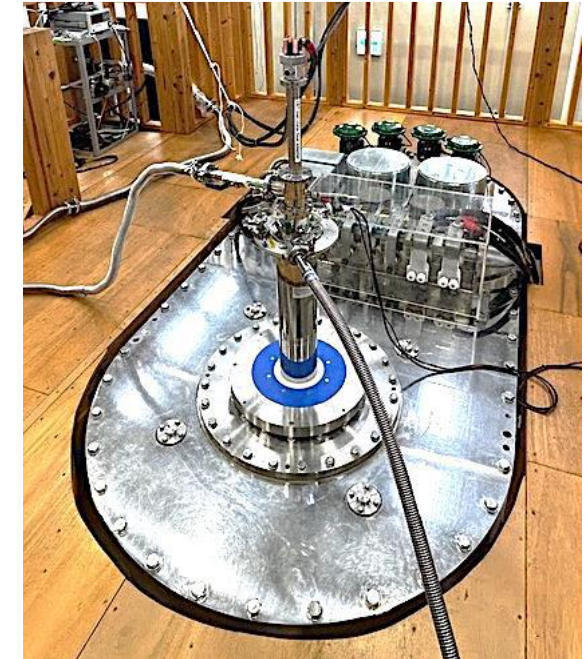


4.1K, 8T and 17T
Nikko, Japan



D. Abraimov *et al*, manuscript
accepted IEEE-TASC

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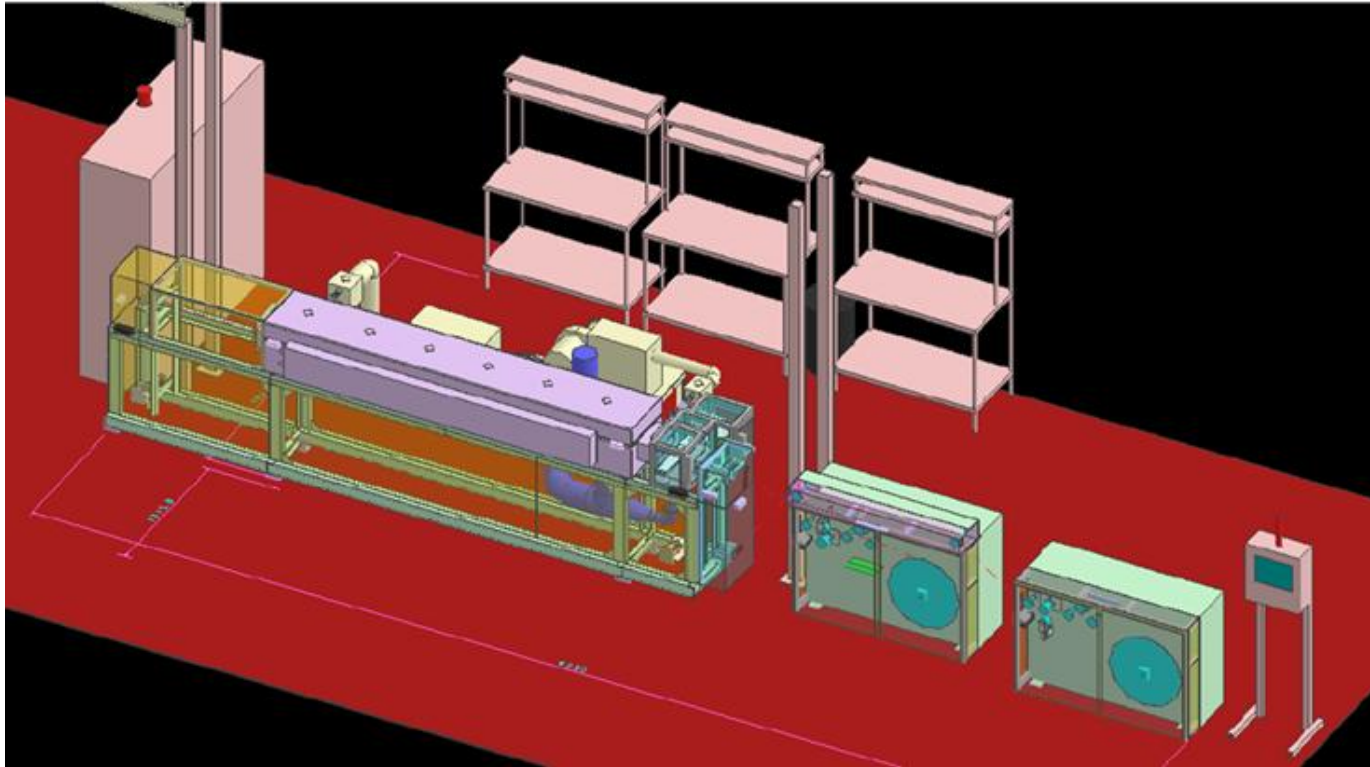
Up to 25 T at various temperatures
Pulsed current up to 5kA
Y. Tsuchiya *et al*, IEEE-TASC
10.1109/TASC.2025.3543148

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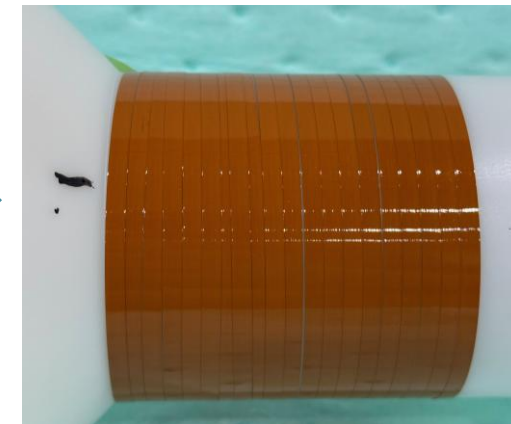


東北大学
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Reel-to-reel polyimide coating system to apply insulation



Currently, insulation is applied by Kapton tape wrapping



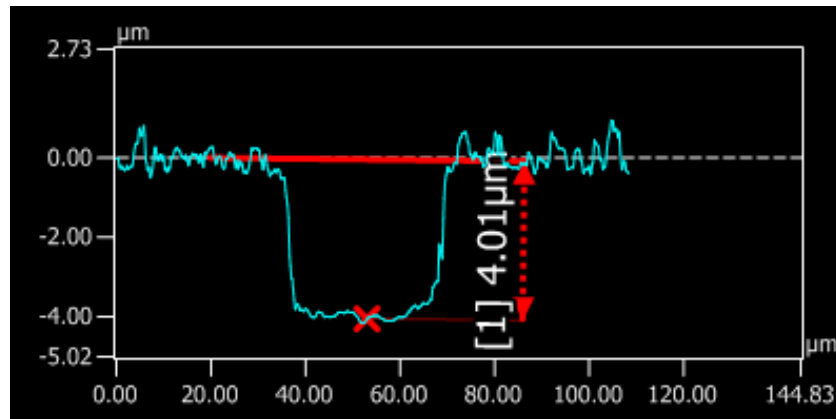
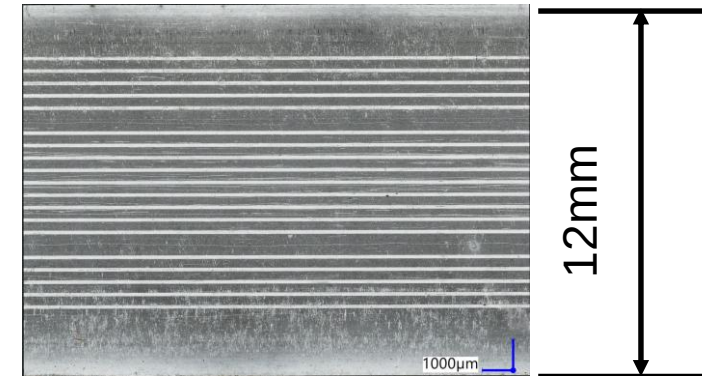
Experimental insulated tape made with PI coating system



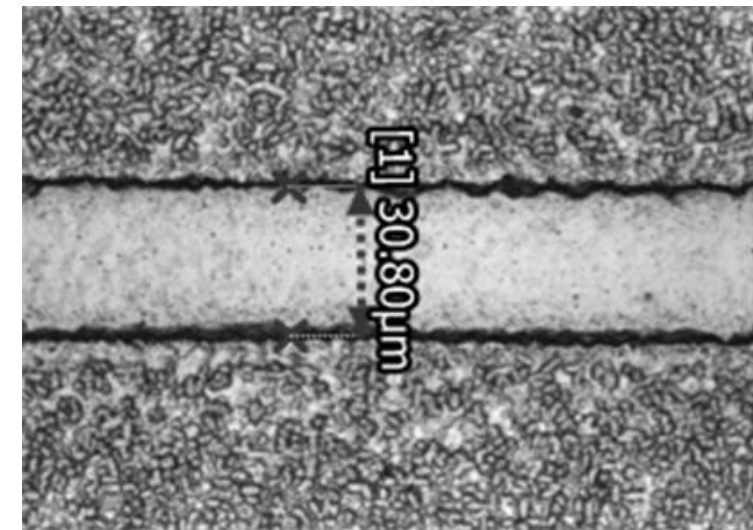
R2R Polyimide coating system to be installed in April 2025

Filamentized REBCO CC via laser striation

- Filamentization performed by laser striation for reduction of AC losses and mitigation of screening current
- Specific striation dimension/geometry, maximizing I_c retention
- Development on capability for long-length processing, compatibility with post-striation processes



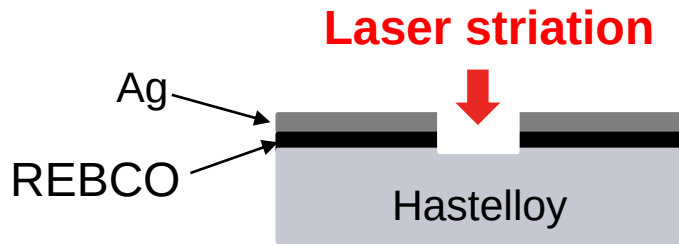
Cross-section depth profile of striation



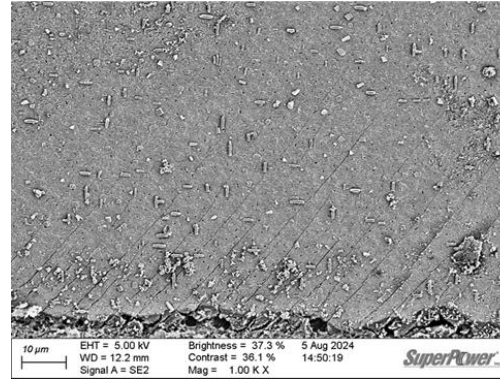
Top view of striation

Development of hybrid slitting to improve slit edge quality

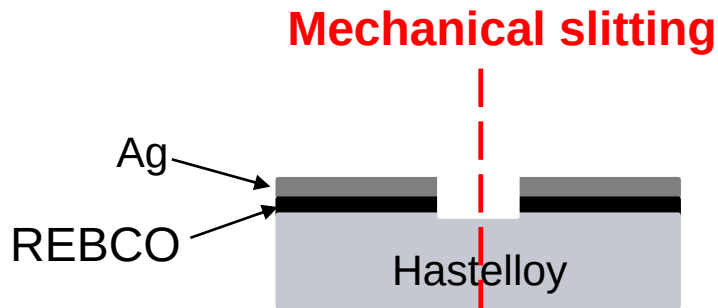
- A two-step process with laser striation prior to mechanical slitting
- Minimizing the damages to the REBCO layer at the slit edges



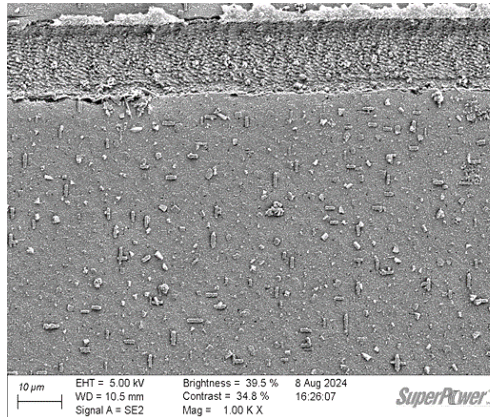
Step 1



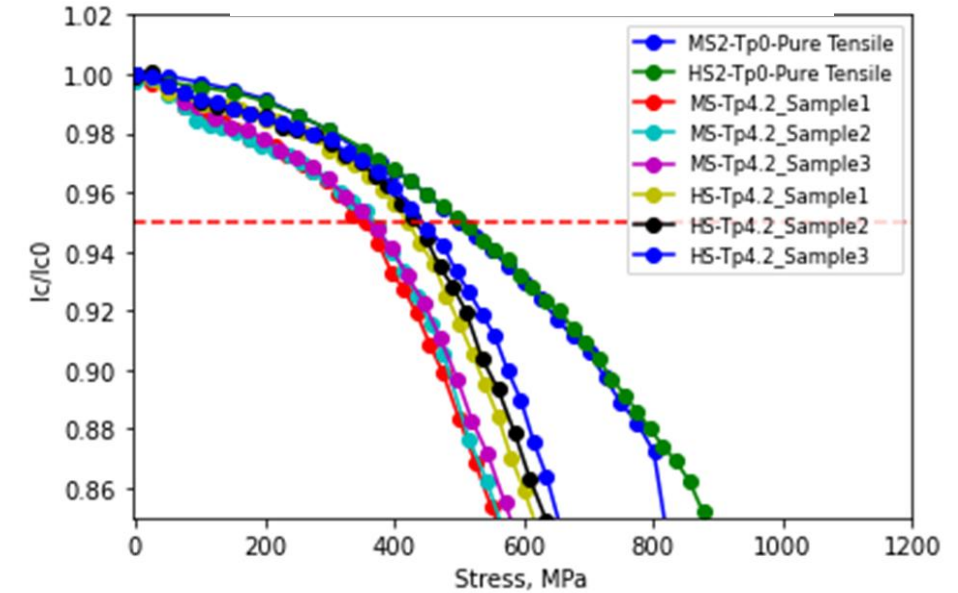
REBCO edge (mechanical slit)



Step 2



REBCO edge (hybrid slit)



Torsion plus tension test showed about 20% increase in critical stress in hybrid slit samples versus mechanical slit samples

Summary

- Manufacturing technology advancement for REBCO HTS wires is driven by the demand and requirements mainly from compact fusion and also from other high-field magnet applications.
- Great efforts are being made to increase yield and throughput, improve processing stability to achieve higher consistency in wire performance and quality, and reduce material cost.
- We are focusing on new REBCO formula development and the design of the next generation manufacturing systems.